

1. Global versus Local Interpretation

Global view	Local view	Consistency
<i>Global SHAP</i> shows that Payload Size and Port dominate the importance distribution across all classes. In the Normal class, <i>Payload Size</i> is the most influential, followed closely by <i>Port</i>. For <i>BotAttack</i> the same two features carry the bulk of influence, while <i>Port</i> is comparatively higher. For <i>PortScan</i>, the contribution of <i>Payload Size</i> is again the largest, but it is the lowest of all classes. <i>LIME</i> rules, being rule-based, also accent the same features, but the weights are smaller in magnitude. The repeated appearance of <i>Port</i> boundaries (e.g., “1.00 < Port 4.00”) and <i>Payload Size</i> thresholds (“Payload_Size > 0.79”) is evidence that the two features dominate the decision surface.	<i>Local SHAP</i> per instance confirms the hierarchy: when a log is predicted as Normal, large positive <i>Payload Size</i> values consistently tilt the decision toward normal traffic; conversely, very negative or very large positive <i>Payload Size</i> values push the sample toward an attack label. <i>Port</i> thresholds (21, 22, 80, 443, 8080) are used in many local explanations to reinforce the global trend.	Highly consistent. The explanations reinforce that both <i>Port</i> and <i>Payload Size</i> are the primary drivers, irrespective of the exact instance.

2. Most Influential Features for “Attack Detected” vs. “Normal”

- The patterns in local explanations show that elevated **Payload Size** (>0.79 in the standardized scale) is strongly associated with an *attack* label, whereas moderated values (0 to 0.5) influence a *normal* classification.
- **Port** number is the second most important predictor. Ports that are domain-specific to services commonly foiled by bots or scanners (e.g.,

22 / SSH, 23 / Telnet, 443 / HTTPS, 8080, 31337) are repeatedly flagged by SHAP and LIME as contributing to the attack prediction.

- Categorical features such as **User_Agent** and **Request_Type** play a supporting role. Agents like *Nikto* (value 1), *nmap* (value 4), or *curl* (value 3) typically carry negative contributions toward a normal outcome (i.e., they bias the decision toward “attack.”) However, the magnitude is small relative to **Port** and **Payload Size**.

3. Patterns & Correlations in the Dataset

- **Port–Payload Size Coupling:** High payload sizes are almost always accompanied by ports that are designated for cryptographic or streaming protocols (e.g., 443, 8080, 443). Such a combination signals potential data exfiltration or port-scanning activity.
- **User_Agent – Attack Signal:** Non-browser agents (*Nikto*, *nmap*, *curl*) appear disproportionately in attack instances. Their presence can be used as a lightweight flag for deeper inspection.
- **Request_Type Frequency:** The majority of normal traffic is HTTP/HTTPS (encoded 2/3), while FTP, SSH, Telnet, SMTP, and Telnet (encoded 5/6) rarely appear in normal logs. When these rarer request types appear, along with a high payload size, the model leans toward an attack prediction.
- **Status Influence:** Successful connections (Status = 1) modestly reinforce normality; failures (Status = 0) tend to tilt the output toward an attack, especially when coupled with high payload sizes or unusual ports.

4. Prediction Quality & Misclassifications

- The accuracy reported (0.9989) and the predictions list contain no misclassifications; every label matches the true label.
- A review of a handful of borderline cases (e.g., a large payload on port 80 with a “curl” user agent) shows that the local SHAP values for these instance are close to zero for most features; the model’s decision is thus highly confident because the pattern is highly unusual for normal traffic.
- Consequently, the mis-classification risk is sparse, but it would most likely surface when an attacker mimics normal traffic—e.g., using a browser user agent to transmit large payloads over port 443.

5. Impact of Categorical Encodings

- The random forest is insensitive to the *ordinal* interpretation of encoded values; a decision tree can split on any threshold without assuming a linear order.
- Nonetheless, the numeric encoding collapses discrete categories into a single numeric space, making it easier for the model to discover thresholds that separate classes (e.g., “User_Agent <= 1.00” effectively isolates

Mozilla vs. attacker tools).

- Should a new category (e.g., a previously unseen agent) appear, the existing encoding would assign it a number that may incorrectly be treated as more similar to a nearby category, potentially diluting the tree splits. One-hot encoding might remedy this at the cost of a slightly larger tree but would guarantee that each category is treated independently.

6. Cybersecurity Insights

- **Payload Size as a Scanner Indicator:** Attack logs have abnormal packet sizes, either heavily inflated or unusually small (reflecting stealth). Firewalls should flag anomalous size patterns, especially on ports that usually carry short requests (e.g., port 22).
- **Agent Fingerprinting:** Agents such as *Nikto* or *nmap* are signals of automated scanning. An early warning system could quarantine traffic from these agents, even if the payload is small, on otherwise benign ports.
- **Port-Based Blocking:** A policy that monitors activity on less common ports (31337, 4444, 6667) and threads a denial-of-service (DoS) log when they appear with large payloads could catch high-risk botnet traffic.
- **Status-Based Filtering:** Failures—especially repeated ones—on critical ports hint at reconnaissance. Combining failure rate with payload size could yield a low-false-positive scanner detection rule.

7. Suggested Improvements

Item	Why & How
One-hot encode categorical features	Avoids ordinal assumptions for <i>User_Agent</i> , <i>Request_Type</i> , <i>Protocol</i> , and <i>Status</i> . Improves interpretability and robustness when new categories appear.
Feature engineering:	

- **Time-between-requests** for a given IP
- **Connection flags** (SYN, ACK) | | These would help the model separate benign long connections from synchronous scans. | | **Introduce a “scan probability” score** |
- Could be the weighted sum of high-confidence attack-class SHAP contributions. | | **Fine-tune class weights** |
- Already balanced, but subtle tweaks may correct rare false negatives. | | **Adopt a deeper tree limit** |
- Reconstruction of tree depth can capture more complex interactions (e.g., high payload + port 22 + “Nikto”). | | **Iterative model evaluation** |
- Compare SHAP vs. LIME on a hold-out; use cross-validation to ensure the patterns hold. | | **Data augmentation** |
- Simulate low-payload bad agents, and high-payload benign browsers to test the model’s discriminatory power. |

The report above synthesizes the SHAP and LIME explanations, identifies the key driving features, links them to real-world cybersecurity heuristics, and offers actionable direction for future model robustness and deployment.